Retail Sales Analysis (January–April 2025)

Project Overview

This project analyzes retail sales data from January to April 2025 using MySQL. The objective is to identify sales trends, top-performing products, revenue-generating categories, and city-wise performance to support business decision-making.

Business Objective

The primary objectives of this project are:

- Analyze overall sales performance.
- Identify the highest revenue-generating products.
- Compare performance across categories.
- Evaluate city-wise sales contribution.
- Measure average order value and transaction trends.
- Generate actionable business insights.

Tools Used

- MySQL
- Microsoft Excel (Dashboard & Visualization)
- GitHub (Project Hosting)

Database Structure

Table Name

Retail Sales Analysis

Columns

Column Name	Data Type	Description
Order_ID	INT	Unique order identifier
Order_Date	DATE	Date of purchase
Product	VARCHAR(50)	Product sold

Column Name	Data Type	Description
Category	VARCHAR(50)	Product category
Quantity	INT	Units sold
Unit_Price	DECIMAL(10,2)	Selling price per unit
City	VARCHAR(50)	Sales location
Revenue	DECIMAL(12,2)	Quantity × Unit Price

Data Preparation

The following steps were performed:

1. Created database and tables.
2. Imported monthly sales data.
3. Calculated Revenue using Quantity × Unit Price.
4. Verified product and category consistency.
5. Combined monthly datasets using UNION ALL for quarterly analysis.

Analysis & Findings

1. Total Revenue Analysis

Business Question

How much revenue was generated during the analysis period?

SQL Concept Used

SUM()

Insight

The business generated significant revenue during the first four months of 2025, indicating strong sales performance.

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2. Monthly Revenue Comparison

Business Question

Which month generated the highest revenue?

SQL Concepts Used

SUM(), GROUP BY, UNION ALL

Insight

Monthly sales trends help identify peak-performing periods and seasonal demand patterns.

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3. Product Performance Analysis

Business Question

Which products generated the highest revenue?

SQL Concepts Used

GROUP BY, SUM(), ORDER BY

Insight

High-ticket products such as laptops contribute a large portion of total revenue despite lower sales volume.

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4. Product Volume Analysis

Business Question

Which products sold the highest number of units?

SQL Concepts Used

SUM(), GROUP BY

Insight

Some products generate high sales volume but contribute less revenue due to lower selling prices.

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5. Category Analysis

Business Question

Which category performs better?

SQL Concepts Used

GROUP BY, SUM()

Insight

Electronics generated the majority of revenue compared to Furniture, indicating stronger customer demand.

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6. Revenue Contribution by Category

Business Question

What percentage of total revenue comes from each category?

SQL Concepts Used

Subqueries, SUM(), GROUP BY

Insight

Understanding category contribution helps management allocate inventory and marketing resources effectively.

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7. City-Wise Revenue Analysis

Business Question

Which city contributes the most revenue?

SQL Concepts Used

GROUP BY, SUM(), ORDER BY

Insight

Top-performing cities can be prioritized for future expansion and promotional campaigns.

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8. Order Value Analysis

Business Question

What is the average order value?

SQL Concepts Used

AVG()

Insight

Average order value helps measure customer spending behavior and sales efficiency.

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9. Top Revenue Orders

Business Question

Which orders generated the highest revenue?

SQL Concepts Used

ORDER BY DESC, LIMIT

Insight

Large-value transactions have a significant impact on overall business performance.

Key Business Insights

- Electronics is the highest revenue-generating category.
- Laptop sales contribute a major portion of total revenue.
- Revenue distribution varies across cities.
- High-revenue products are not always the highest-volume products.
- Average order value provides insight into customer purchasing behavior.

Conclusion

This project demonstrates the use of MySQL for retail sales analysis through data preparation, querying, aggregation, and business reporting. The analysis provides valuable insights into product performance, category contribution, city-wise sales trends, and overall business growth.

Future enhancements include creating interactive dashboards in Excel and Power BI for visualization and executive reporting.